

Classical trajectory method. Quantum Entanglement from Classical Trajectories

Today I will talk about the classical trajectories method.

We use this method to simulate large molecules or solids. The idea is simple: nuclei move by Newton's laws (like small balls), and electrons are treated quantum mechanically.

Example: sodium iodide molecule (NaI).

In the ground state, the molecule is Na and I. Nuclei move classically. The laser gives energy to the electron, and it jumps to another state. Now the molecule is Na I. It starts to fly apart.

Near the point where the two energy levels cross, the electron can jump back or stay. Two outcomes are possible:

ions: $\text{Na}^+ + \text{I}^-$

neutral atoms: $\text{Na} + \text{I}$

A good classical trajectories method must show this splitting (branching).